

AIVOA.AI – Round 1 AI Product Engineer (Fresher)

AI-Powered Customer Complaint Management System

Company Website: <https://aivoa.ai>

Objective

Build an **AI-powered Customer Complaint Management System** for the pharmaceutical manufacturing industry using the **reference UI** and **demo video** provided.

Before You Start

Before implementation, please:

- Watch the demo video completely.
- Read this assignment document.
- Research **Quality Management System (QMS)** used by pharmaceutical companies manufacturing **API** and **FDF**.
- Understand the purpose of the **Customer Complaint** module in a QMS.

We are not looking for domain experts. We are looking for curiosity, research, and problem-solving.

Mandatory Technology Stack

- Frontend: **React** UI with **Redux** for state management.
- Backend: **Python** with **FastAPI**.
- **AI Agent Framework: LangGraph.**

- LLMs: Utilize Groq(<https://console.groq.com/docs/models>), specifically **the gemma2-9b-it model** (assume you will create a new API token for this). You may also consider **llama-3.3-70b-versatile** for context.
 - Database: MySQL/Postgres SQL
 - Font: Google Inter.
 - You may use Gemini 2.5 Pro (1-month free trial) or ChatGPT 5.0 for all coding and research, with zero human-written code.
Note: Do not simply copy and paste AI-generated code. Understand what the code does, then implement and adapt it to match the workflow shown in the demo.
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Reference Material

Demo Video:

https://drive.google.com/file/d/1av2lzDPx8YMSzTrlz7w51HTRWBz3_5Nj/view?usp=sharing

Reference UI Screenshot

Log Customer Complaint

API & FDF Quality Assurance Module

Pending Triage

AI Complaint Intake Assistant

BETA

Drag & drop complaint document here
or [click to browse](#)

OR

Paste Complaint Text / Email

Supported formats: PDF, DOCX, TXT, EML
Max file size: 10MB

EXTRACTION PROGRESS

10%

Analyzing document content and extracting key details...
Please wait, this may take a few moments.

AI ASSISTANT

Upload a complaint document or paste text above.
I will automatically extract the details and populate the form for you.

Ask me anything about this complaint...

AI responses may contain errors. Please verify information.

1. ORIGIN & CUSTOMER DETAILS

Complaint Source

Awaiting AI extraction...

Customer Name

Awaiting AI extraction...

2. PRODUCT & BATCH IDENTIFICATION

Product Name

Awaiting AI extraction...

Product Strength/Grade

Awaiting AI extraction...

Batch/Lot Number

Awaiting AI extraction...

Manufacturing Date

Awaiting AI extraction...

Expiry Date

Awaiting AI extraction...

Quantity Affected

Awaiting AI extraction...

kg

3. COMPLAINT DETAILS

Complaint Type

Awaiting AI extraction...

Complaint Date

Awaiting AI extraction...

Detailed Complaint Description

Awaiting AI extraction...

4. INITIAL ASSESSMENT & PRIORITY

Initial Severity

Awaiting AI extraction...

Priority

Awaiting AI extraction...

Reset Form

Save Complaint

Implementation

- Follow the workflow demonstrated in the demo video.
- Use the mandatory technology stack.
- The UI doesn't need to match the screenshot exactly, but the demonstrated functionality should be implemented.
- You may create your own realistic pharmaceutical complaint PDFs, emails, or images for demonstration.
- Production-grade OCR or document parsing is **not required**.

Bonus Features *(Optional)*

Additional AI features are highly appreciated.

Examples:

- Complaint Completeness Checker
 - Root Cause Recommendation
 - Duplicate Complaint Detection
 - CAPA Recommendation
 - Complaint Summary
 - AI Risk Classification
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Deliverables

Submit your assignment using the link below:

Submission Form:

<https://forms.gle/JhB65RNgSzHyBrBm7>

Include:

- GitHub Repository
 - **5–10 minutes demo video** explaining: **(2 video submission)**
 - **Working demonstration** of all implemented AI tools and frontend features.
 - **Demonstrate and explain the code** by walking through the complete end-to-end workflow, starting from the user's input (prompt or PDF/email upload) in the frontend, showing the relevant frontend code, API endpoints, backend processing, AI/LangGraph workflow, and finally how the response populates the **Log Customer Complaint** form and **AI Copilot Risk Assessment**.
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Important Notes

- AI tools such as **ChatGPT, Gemini, Claude, GitHub Copilot, Cursor, Windsurf**, or similar tools are **allowed and encouraged**.
- **Do not simply copy and paste AI-generated code.** Understand what the code does, then implement and adapt it to match the workflow shown in the demo.
- During the interview, you should be able to explain your implementation and may be asked to modify or extend your solution.
- We value **curiosity, clean code, product thinking, and problem-solving** more than building a perfect application.

Good Luck!

We look forward to seeing your creativity and engineering skills.

Team AIVOA.AI